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Total No. of Questions : 8] *[Total No. of Printed Pages : 1*

Roll No

EE-402 (GS)

B.E. IV Semester Examination, June 2020

Grading System (GS)

Electromagnetic Theory

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

l. Define Gaussian surface. Calculate total electric flux leaving the cylindrical surface $\rho = 4.5$, $z = \pm 3.5$ for a line charge on x-axis with $\rho_L = 2 \cos (0.1x)$, x is in radians.

2. Derive the expression for transmissions coefficient and reflection coefficient of uniform plane waves with normal incidence at plane dielectric boundary.

3. a) State and explain Ampere's circuital law.
b) What do you understand by boundary conditions? Explain boundary conditions due to magnetic field.

4. Write short notes on :
i) Poisson's equation ii) Uniqueness theorem

5. a) What is polarization? Explain polarization of waves and their types.
b) Deduce Maxwell's equation in differential form by preparing tabular chart.

6. Find the magnitude of electric field intensity in a simple of silver having $\sigma = 6.17 \times 10^7$ mho/m and $\mu_e = 0.0056$ m^2/V–S. If
i) Drift velocity is 1 mm/s ii) Current density in 10^7 A/m.

OR

Define electrostatic potential and electric field intensity. Hence show that, $E = -\nabla V$.

7. Obtain electric field in all regions due to following charge distribution in free space.

$$\rho (r, \phi, z) = 0 , 0 < r < a$$
$$= \rho_0 , a < r < b$$
$$= 0 , b < r < \infty$$

OR

For uniform plane wave in sea water, $\sigma = 4$ S/mt, $\in = 80 \in_0$, $\mu = \mu_0$. Find α, β, η and λ at
i) 10,000mHz
ii) 25 KHz

8. Write short notes on: (any two)
a) Method of images
b) Magnetic dipole and its moment
c) Biot-Savart law
d) Divergence theorem
e) Electric field intensity

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